

Abhishek kumar XR Developer

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Education

1.B-tech in Mechanical Engineering, Greater Noida Institute of Technology

08/2021 – 06/2024 | Greater Noida, India

Skills

XR Development | Unity | Vuforia Engine | Mechanical Design | Autocad | Solidworks | Fusion 360 | Blender | ANSA | Digital sketching | Photoshop | C Programming | C# | Effective Team Leader - Communication, Decision Making, Team Player

Professional Experience

AR Learning Intern, IIT Delhi AIA Foundation for Smart Manufacturing

06/2023 – 07/2023 | Delhi, India

- Developed an AR-based simulation using Unity and Vuforia Engine, utilizing image target tracking to overlay the MPRC CAD model and enable interactive process visualization.
- Integrated UI controls for real-time MPRC simulation, allowing users to demonstrate and analyze operations from multiple perspectives on mobile

VR Learning Intern, IIT Delhi AIA Foundation for Smart Manufacturing

02/2024 – 07/2024 | Delhi, India

- Developed a VR-based training application called RWC VR using Unity along with the XR Interaction Toolkit to enable immersive interaction experiences.
- Designed interactive training modules covering hardware introduction, startup sequence, and welding process simulation, enhancing hands-on learning in a virtual environment.

XR Developer, IIT Delhi AIA Foundation for Smart Manufacturing

08/2024 – present | Delhi, India

- Built a VR-based pneumatic training module using Unity and XR Interaction Toolkit, offering an immersive hands-on experience.
- Enabled users to virtually assemble and connect pneumatic components like Single-Acting Cylinders, Double-Acting Cylinders, and Electro-Pneumatic Systems for practical learning.

AR/VR Introduction Course – Developer & Instructor, IIT Delhi AIA Foundation for Smart Manufacturing

10/2024 – 11/2024

- Created a beginner-friendly course covering the fundamentals of AR/VR using Unity. Trained learners to build and deploy functional AR/VR applications independently. Each lecture was followed by quizzes and assignments to reinforce learning and ensure hands-on practice.

Projects

FSM Lab VR Tour for IITD AIA FSM

05/2024 – 07/2024

- Developed a VR-based lab tour covering CPL, CPTF, and CPF facilities, allowing users to explore FSM's infrastructure remotely.
- Integrated interactive touchpoints for machine insights and simulations to demonstrate operational workflows in a virtual environment.

Autonomous Mobile Robot (AMR) Design for UN World Food Programme

11/2024 – 12/2025

- Designed an Autonomous Mobile Robot (AMR) using Blender and Fusion 360 for warehouse navigation and real-time food monitoring.
- AMR includes sensors to collect the warehouse environment live data, reflecting on a dashboard and triggering alerts for anomalies.

VR-Based Training Application for Vehicle Gear Assembly – Maruti Vendor

01/2025 – 03/2025

- Developed an immersive VR-based training application for an automotive assembly line focused on the vehicle gear mechanism. The project consisted of three core modules:
 - Awareness Room** – A virtual environment simulating the assembly line where users interact with UI elements to learn about station hardware and visualise component functionality through simulations.
 - Training Room** – A guided hands-on training space featuring an instruction panel and checklist. Users perform the gear assembly process with step-by-step interactive guidance, ensuring adherence to standardised procedures.
 - Evaluation Room** – A self-assessment environment where users perform tasks independently with optional contextual hints. Based on task accuracy, users receive a Pass/Fail status. Performance data is recorded and a report card is generated on a connected web portal.

VR-Based Training Application for Vehicle Headlight Assembly – Maruti Vendor

04/2025 – 05/2025

- Designed and developed a bilingual (English & Hindi) VR training application for an automotive Headlight assembly line. The application features:
 - Awareness Room:** Interactive simulation to explore station hardware and understand component functions.
 - Training Room:** Guided, step-by-step assembly tasks with real-time instructions and a progress checklist.
 - Evaluation Room:** Self-paced assessment with contextual hints; generates a Pass/Fail result and uploads user performance to a connected web portal.

Maintenance through AR, AR-Based Industrial Troubleshooting Application

- Designed an AR app for troubleshooting servo motors, RFID, and proximity sensors in industrial systems.
- Enabled interactive checks for wiring, power supply, and loose connections.
- Provided guided replacement procedures to minimize downtime.

SmartLabXR, Mixed Reality Smart Manufacturing Laboratory Experience

- Developed an immersive MR smart manufacturing laboratory experience in Unity, enabling users to explore physical machines through spatially aligned digital content.
- Implemented spatial tracking, LiDAR-based environment understanding, interactive 3D machine visualization, and indoor navigation for the laboratory.
- Integrated IIoT data visualization to present real-time machine information directly within the physical environment.

VMC AR, AR-Based VMC Operator Training

- Developed an **AR-based VMC operator training application** using Unity and Vuforia Model Target tracking.
- Implemented physical **VMC recognition and spatially aligned instructional content** over the machine control panel.
- Developed step-by-step training modules for **spindle control, coolant control, tool change, and tool offset operations**.
- Created contextual visual guidance to help operators learn and execute individual **VMC machine-control procedures**.

Pneumatic Trainer VR - Version 2, Interactive VR Pneumatic Circuit Training System

- Developed an interactive **VR pneumatic trainer** in Unity for virtual assembly and operation of pneumatic circuits.
- Implemented **grab-and-place components, pneumatic hose connections, valve interactions, circuit configuration, and pressure visualization**.
- Simulated **Single-Acting Cylinder, Double-Acting Cylinder, 3/2 & 5/2 directional valves, flow-control valves, and electro-pneumatic concepts**.
- Developed a pneumatic network and visualization system allowing users to **construct, operate, modify, and experiment with virtual pneumatic circuits**. Pasted text

Note: If you're trying to make the CV more focused on your current XR career, I would probably **remove this project from the final CV** and keep it on your portfolio.

XR Trainer

Workshops Delivered for Maruti Suzuki

Introduction to XR – Workshop

- Trained Maruti Suzuki personnel on XR concepts, ecosystem, and potential use in manufacturing and training.

Maintenance through AR – Hands-on Workshop

- Led a practical AR session where participants developed a machine-maintenance AR application, demonstrating step-by-step maintenance procedures using AR.

Workshops Delivered for Tata Motors

Introduction to XR – Workshop

- Trained Maruti Suzuki personnel on XR concepts, ecosystem, and potential use in manufacturing and training.

Maintenance through AR – Hands-on Workshop

- Led a practical AR session where participants developed a machine-maintenance AR application, demonstrating step-by-step maintenance procedures using AR

Honors and Awards

- **IIMT infinity fest** - Cash prize and Certificate winner in AUTOCAD Drafting.
09/12/2022
- **XRDC Design Challenge** - Gained hands-on experience in the latest AI integration with XR
05/01/2025
- **XRCC Hachathon** - Created a Smart Lab XR project which aims to demonstrate the lab facilities immersively using MR based application.
06/05/2026

Certificates

Complete C# Unity Game Developer 3D (Updated To Unity 6) | Unity C# Scripting : Complete C# For Unity Game Development. |
IITD : Smart Manufacturing in XR | CETPA : Technical Drawing in AUTOCAD | CETPA : CAD Modelling in SOLIDWORKS. |